Date: Assignment#3 Day:

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Digital Logic Design
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SE-151
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Q1 What is a Finite State Machine (FSM)? Explain the difference between sequential and combinational circuits. Also Explain the basic components of an FSM.

A A Finite State Machine is a mathematical model of computation used to design both, Computer Programs and Sequential Logic Circuits. It is concieved as an abstract machine that can be in exactly one of a finite number of states at any given time, transitioning between them based on inputs.

Combinational Circuit: The output depends only on the current inputs present at that exact moment. They have no memory of past inputs. Its examples are MUXs, Adders etc.
Sequential Circuit: The output depends on both, the current input and the past history of inputs. They require memory elements to store the current state. Its examples are Flip Flops, Register etc.

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The basic components of FSM are:

State: Represents a specific condition or situation of the system based on its history.

Input: The external signal or data that is fed into machine and triggers transitions.

Output: The response that is generated by the machine.

Clock: A timing signal used in Synchronous FSMs to coordinate state transitions at discrete intervals.

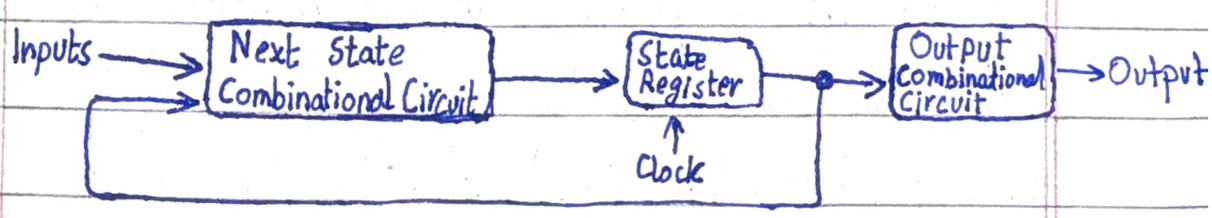
Memory Element: Hardware components (like flip-flops) used to store the binary representation of current state.

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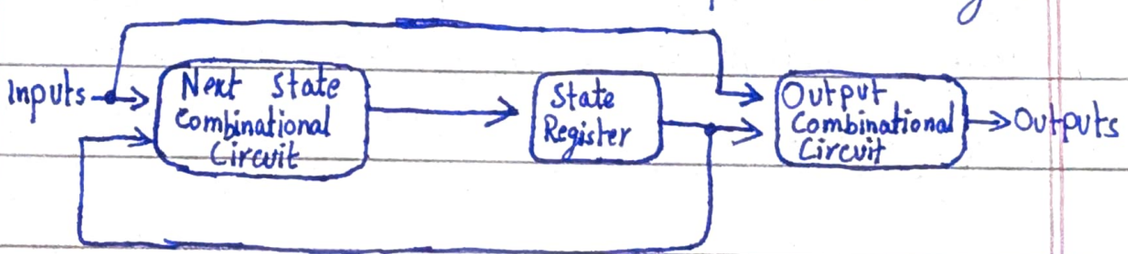
Q2 Explain Moore Machine and Mealey Machine with diagrams. On what factor does the output depends in both machines? Also Explain:

- Advantages
- Disadvantages
- Real-life Applications

A Moore Machine: The output depends only on the current state in this system.



Mealey Machine: The output depends on both the current state and current input in this system.



As we can see from above diagrams that

- Moore has no direct wire from input to output therefore, output depends on current state only.
- Mealey has a direct wire from input to output which proves that output depends on current state and current input

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Advantages & Disadvantages

Advantages:

Moore	Mealey
• Outputs are more stable and synchronous with clock	Reaction times are faster.
• The chance of output glitching is reduced	Generally requires fewer states

Disadvantages:

Moore	Mealey
• ~~Fa~~ Lower Reaction times	Outputs are prone to glitching if inputs change mid-cycle.
• Generally requires more states	

Real-Life Applications:-

- Traffic Light Controllers
- Vending Machines
- Combination Locks

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Q3. Write a detailed comparison between Moore and Mealey Machine. Compare:

- Output Dependence
- Number of states
- Speed
- Stability
- Hardware Complexity

A Output Dependence:-

Moore Machine depends solely on current state.
Mealey Machine depends on both the current state and the input.

Number of States:-

Moore Machine generally requires more states.
Mealey Machine generally requires fewer states

Speed:-

Moore Machine is generally slower; Outputs are delayed by a clock cycle.
Mealey Machine is generally faster; Outputs change immediately with the input.

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Stability:-

Moore Machine is highly stable and immune to input glitches mid-cycle.

Mealey Machine is less stable and more prone to input glitches leading to false outputs.

Hardware ~~Compatibility~~: Complexity

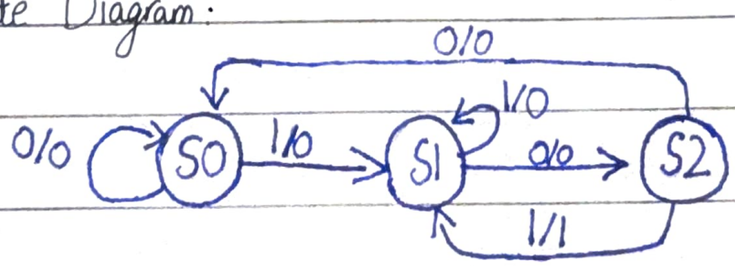
Moore Machine requires more states however the output logic is simpler.

Mealey Machine requires fewer states but its combinational output logic is more complex.

Q Design the state diagram, state table and output table of an FSM that detects binary sequence "101".

FSM for binary sequence "101". Assuming Mealey machine

State Diagram:



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State Table:

Current State	Next State (Input=0)	Next State (Input=1)
S0	S0	S1
S1	S2	S1
S2	S0	S1

Output Table:

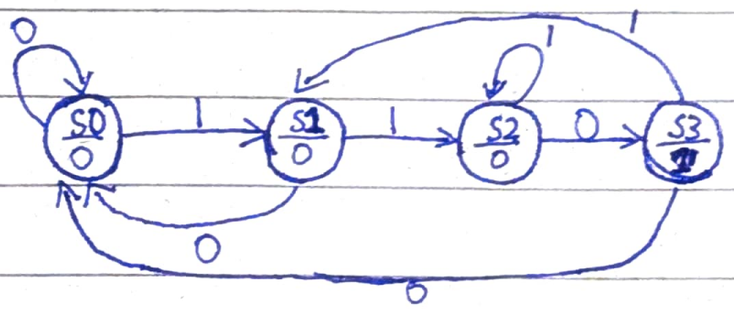
Current State	Output (Input=0)	Output (Input=1)
S0	0	0
S1	0	0
S2	0	1

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Q5 Design a Moore Machine sequence detector for binary sequence "110".

- State diagram
- State Table
- State Equations
- Explain why is it a Moore Machine

State Diagram:-



State Table :-

Current State	Next State (Input=0)	Next State (Input=1)	Output
S0	S0	S1	0
S1	S0	S2	0
S2	S3	S2	0
S3	S0	S1	1

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Output Equations:-

Q_0 \ Q_1	0	1
0	0	0
1	0	(1)

$Z = Q_0 . Q_1$

Why is it a Moore machine:-

The output column in the table depends entirely on "Current State" and is entirely independent of input variable.

Q Explain the procedure for the design of a Mealy Machine sequence detector for sequence "1011".

- State Transitions
- State Diagram
- State Table
- Output Conditions
- Explain Why is it a mealy machine

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A State Transition:-

From S0:-

If input is 0, Stay in S0, Output is 0

If input is 1, Move to S1, Output is 0

From S1:-

If input is 0, ~~Stay~~ Move to S2, Output is 0

If input is 1, Stay in S1, Output is 0

From S2:-

If input is 0, Go back to S0, Output is 0

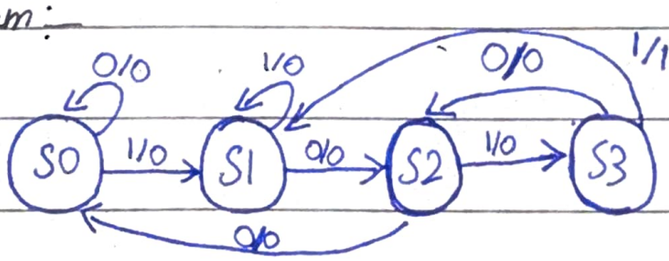
If input is 1, Move to S3, Output is 0

From S3:-

If input is 0, Go back to S2, Output is 0

If input is 1, Move to S1, Output is 1

State Diagram:-



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State Table:-

Current State	Next State(Input=0)	Next State(Input=1)	Output(Input=0)	Output(Input=1)
S0	S0	S1	0	0
S1	S2	S1	0	0
S2	S0	S3	0	0
S3	S2	S1	0	1

Output Conditions:-

The output equation will only equal 1 if:

- The machine must currenly be sitting in S3
- The present input must be 1.

If any of condition is False, the output remains 0.

Explain why its mealy machine:-

The design is mealy because the final output of S1 is not tied exclusively to S3. The machine must simultaneously look at the state and current input to generate the output. So, because the output depends on both the state and input, It can be described as a mealy machine.

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Q5 Explain the procedure for converting a mealey machine into a moore machine and a moore machine into mealey machine with suitable examples: Include:

- Conversion Rules
- State Splitting
- Advantages of Conversion
- Example State Diagrams

Converting a Mealy Machine to Moore Machine :-

For the conversion of mealey machine to moore machine, We choose a sequence detector for binary ~~detector~~ Sequence "101".

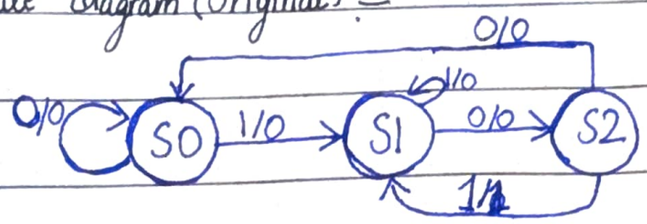
Conversion Rules :-

1) Look at incoming arrows for every mealy state.
2) If all incoming arrows to a state share the same output, move that output inside the state to make it a moore state.
3) If the incoming arrows have conflicting outputs, you must split state into two states; One for output "0" and one for output "1".
4) Route the incoming arrows to newly split states that matches their specific output.

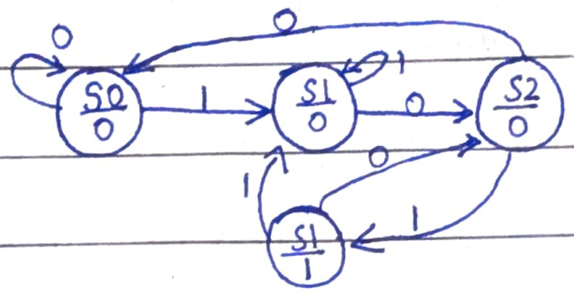
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5) Copy all original outgoing paths to both of newly split states.

State diagram (Original):-



State diagram (Converted):-



State Splitting:-

As seen in the example above, S1 recieved 2 different outputs, therefore, we split it into 2 new states.

Why Convert Mealy to Moore:-

Stability & Glitch Elimination:

Mealy Machine outputs can fluctuate wildly if input signals is noisy or change before clock ticks. By converting to Moore Machine, the output is strictly tied to state register, meaning the output is synchronized with clock-cycle and immune to mid-cycle input glitches.

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Safer Hardware: Systems like traffic lights or medical equipment prefer moore designs because output logic is simpler and determined based on current state.

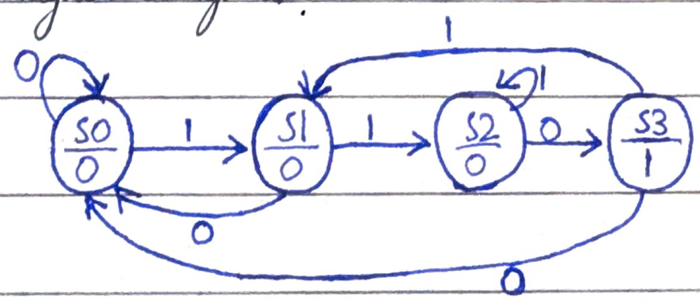
Moore Machine to Mealey Machine:-

For this conversion, we pick a sequence detector for binary sequence "110".

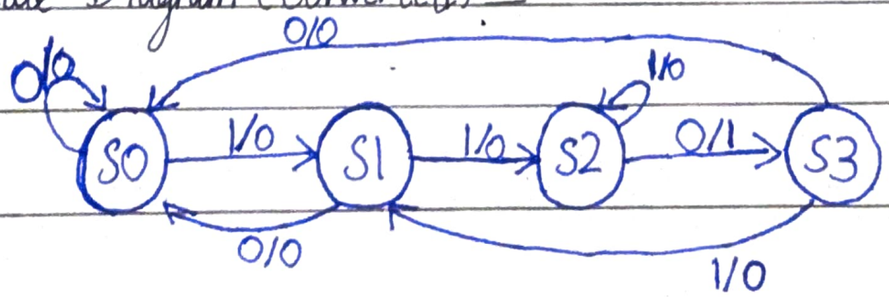
Conversion Rules:-

1) Identify the output associated with each Moore state.
2) Remove the output from inside the state.
3) Attatch that output value to every incoming transition edge that points to that specific state.

State Diagram (Original):-



State Diagram (Converted):-



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Why Convert Moore to Mealy:-

Fewer States: As we saw in examples above, Converting to Moore forced us to add an extra state, Converting back to Mealey would have allowed us to compress states, which would lead to fewer Flip-flops thus saving space and costs.

Speed: A Moore Machine has a built in one-clock-cycle delay because input must change the state, and then state changes the output. A mealey machine bypasses this, the moment input hits, the output updates instantly in that exact same clock style.